

MEHAR BABA COMPETITION-3



**Mehar Baba
Competition** 3
Collaborative Drone Based
Surveillance Radars

**DIRECTORATE OF OPERATIONS (REMOTE)
AIR HEADQUARTER (VAYU BHAWAN)**

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TABLE OF CONTENTS

1. Background
 - 1.1. Mehar Baba Competition
 - 1.2. Swarming of Unmanned Aerial Systems
 - 1.3. Collaborative Drone Based Surveillance Radar
 - 1.4. Relevance of Concept
2. Important Dates
3. Competition Description
 - 3.1. Vision
 - 3.2. Structure and Specifications
 - 3.3. Broad Parameters
4. Eligibility
5. Innovation
6. Intellectual Property
7. Application and Submission of Information
8. Evaluation Information
 - 8.1. Committee of Experts
 - 8.2. Evaluation Criteria
9. Award Administration Information
10. Terms and Conditions for Participation
11. Competition Contacts

1. Background

1.1. Mehar Baba Competition. Named after Air Commodore Mehar Singh DSO MVC, the Mehar Baba Competition (MBC) is an IAF initiative to encourage indigenous development of niche technology. The competition was launched in 2018 and it promotes innovation among Indian industries and academia. After the successful culmination of two editions of the competition, MBC-3 was announced in Feb 2025 with the theme 'Collaborative Drone Based Surveillance Radar'. This edition envisages development of a Proof of Concept for a collaborative solution with a swarm of Unmanned Aerial Systems (UAS) working as a distributed airborne radar capable of detecting, tracking and reporting aerial targets in a contested environment.

1.2. Swarming of Unmanned Aerial Systems. Unmanned Aerial Systems (UAS) are transforming the aerospace industry by providing accessible and relatively inexpensive aerial platforms for an extremely wide variety of applications on a scale not realizable with piloted aircraft. Swarm intelligence is inspired by natural systems such as flocks of birds where collective behaviour enables multiple UAS to operate as an efficient cohesive unit, communicating and collaborating autonomously to complete complex missions. By incorporating swarm intelligence into existing UAS technologies, their capabilities can be significantly enhanced towards collaborative decision making, autonomous threat detection and efficient mission execution.

1.3. Collaborative Drone Based Surveillance Radar. This edition of MBC envisages development of a Proof of Concept for a collaborative swarm of Unmanned Aerial Systems (UAS) working as an airborne radar capable of detecting, tracking and reporting aerial targets in a contested environment and provide their accurate location at a centralised monitoring station. The collaborative nature facilitates operational continuity even if individual UAS are compromised, while their scalability allows them to adapt to missions of varying complexity, seamlessly integrated with existing surveillance systems to provide real time intelligence for enhanced situational awareness and effective decision making.

1.4. Relevance of Concept. This capability has a big use for conventional air operations, primarily by reducing risks to an AWACS class of platform, increasing mission survivability and enhancing the mission capabilities for e.g. pushing the distributed surveillance radar system into the enemy air space rather than operating stand-off. However, a development at that scale is not possible within the budget of MBC. Therefore, this Proof of Concept is being done at a smaller scale as a C-UAS capability. This capability itself has great potential given the current weakness of C-UAS systems in terms of detection. The spin-offs from the Proof of Concept will flow both into the C-UAS mission against drones and at a later date into a distributed AWACS concept.

2. Important Dates

Event	Date
Competition Registration Commencement	27 April 2026
Application Deadline	31 May 2026, 2355 Hr
Phase 0: Shortlisting of Applications	02 - 12 June 2026
Phase I: Declaration of Shortlisted Participants	18 June 2026
Phase I: Presentations	13 - 24 July 2026
Phase I: Result	14 Aug 2026
Phase II: Evaluation of Limited Demonstration	To be announced
Phase II: Result	To be announced
Phase III: Industrial and Capability Visits (Progress Checks)	To be announced
Phase IV: Evaluation of Full Demonstration	To be announced
Phase IV: Result	To be announced
Final Award Ceremony	To be announced

Note: Please visit the IAF website (https://indianairforce.nic.in/mehar_baba/) for updates on dates and events.

3. Competition Description

3.1. Vision. To stimulate indigenous development of Unmanned Aerial Systems (UAS), MBC-3 aims to develop the Proof of Concept for a collaborative solution with a swarm of Unmanned Aerial Systems (UAS) working as an airborne radar capable of detecting, tracking and reporting aerial targets in a contested environment and provide their accurate location at a centralised monitoring station. By operating autonomously and dispersing across predefined sectors, swarm based airborne radar promises to provide extensive area coverage while collaboratively relaying real time data. The distributed nature of the system has the potential to ensure mission survivability and redundancy.

3.2. Structure and Specifications. The competition shall span over five mandatory phases with progressive elimination based on an evaluation criterion for each phase. The IAF shall also provide financial support for research, development and capability demonstrations in Phase-II, Phase-III and Phase-IV. The details are mentioned below:-

3.2.1. Phase-0. Initial application submission and shortlisting of up to 30 participants based on technical and operational proposals.

3.2.2. Phase-I. Presentations by shortlisted participants at a specified date and time at a designated venue at New Delhi. Video links for demonstrating capabilities, functionalities, available hardware, etc. are encouraged during this presentation. Simulation of utilised algorithms to demonstrate the proposed solution are also encouraged. Participants are free to use any known standard/ open source simulator

to show various functionalities and capabilities. Up to ten participants shall qualify at the end of Phase-I to advance into the next phase of the competition.

3.2.3. Phase-II. Demonstration of limited capability for collaborative swarm based airborne radar for detecting targets based on a specified evaluation criterion. The demonstration shall be held for all ten participants at a designated place and time as per the finalised schedule. Up to five finalists shall be selected at the end of Phase-II.

3.2.4. Phase-III. Industrial visits by members of Committee of Experts to the facilities of the finalists to assess their manufacturing and integration capabilities, ensuring scalability and readiness for indigenous production.

3.2.5. Phase-IV. Full demonstration of Collaborative Drone Based Surveillance Radar with adherence to technical specifications and operational requirements as mentioned in the Appendix to this document. The assessment shall be done by a nominated Committee of Experts as per the pre-defined evaluation criterion. Three winners will be announced at the end of this phase.

3.3. Broad Parameters. The details of broad parameters including technical specifications, operational requirements and desirable features are mentioned in the Appendix.

4. Eligibility

4.1. The competition is open for Indian citizens and Indian registered entities only. Indian registered for-profit companies, entities, academic institutions and teams of Indian citizens are eligible to participate in the competition. This competition is not open to any air warrior from the sponsoring cell.

4.2. Groups of Indian citizens, e.g. college students, without a formal registered for-profit entity or start-up can participate only in Phase-0 and Phase-I. However, upon selection for participation in Phase-II and subsequent continued participation in the competition, the group shall need to form an established for-profit Indian company. No exceptions shall be allowed.

4.3. The IAF also encourages collaboration or joint participation through team mergers or joint venturing in case the jointly participating teams conclude that their combined skills shall enhance the product functionalities and increase their chances of success in the competition. Each participant/ team can participate in only one such collaboration.

5. Innovation

5.1. The participants are free to come up with any innovative or bold solution within the bounds of this Vision Document. Plagiarism of ideas, theft of Intellectual Property or copying of solutions is strictly prohibited. Participants are encouraged to maintain references for technologies used to work out the solutions.

5.2. Primarily, innovation is expected and encouraged towards swarm coordination, application of AI, airborne radar technology, threat analysis, decision making, providing alerts and suggesting corrective measures.

6. **Intellectual Property.** All Intellectual Property (IP) created during the competition shall be jointly owned by the developing team and the IAF. The developing team and the IAF will get into a binding contract of co-ownership at a later date.

7. **Application and Submission Information**

7.1. Interested participants are to submit their application by filling requisite details in the form available on the webpage of the competition. A write-up of approximately 500 words is also to be submitted as per the specified template covering details of the plan. The proposal should focus on meeting the requirements of the competition.

7.2. Non-compliance with the template while submission of application shall result in disqualification.

7.3. Interested applicants are advised to read all sections of this Vision Document carefully before applying.

8. **Evaluation Information**

8.1. **Committee of Experts.** Each phase shall be evaluated by a pre-selected panel of Committee of Experts. The details of the Committee shall remain confidential. Each phase may not have same composition, but shall comprise members with similar cumulative skills, expertise and backgrounds. The Committee may include members from following backgrounds:-

8.1.1. Serving officers with relevant background and/ or operational experience.

8.1.2. Domain expert(s) from R & D Organisations/ Civil Organisations/ Academia.

8.1.3. Co-opting of other domain experts on case to case and/ or need basis for different phases.

8.2. **Evaluation Criteria.** Along with the operational goals, the collaborative UAS solution will need to demonstrate the following technical abilities at the various phases of the competition. The participants/ teams shall be adjudged by the Committee of Experts (CoE) on multiple levels including but not limited to following characteristics/ features:-

8.2.1. **Radar Characteristics for Swarm**

8.2.1.1. Method proposed/ Type of radar

8.2.1.2. Detection envelope

- 8.2.1.3. Target characteristics: RCS, Velocity, etc.
- 8.2.1.4. Operational time of radar
- 8.2.1.5. Tracking ability
- 8.2.1.6. Accuracy and resolution
- 8.2.1.7. Detection and tracking of multiple targets
- 8.2.1.8. Detect and track while the swarm is moving
- 8.2.1.9. Graceful degradation

8.2.2. Collaborative Swarm Behaviour

- 8.2.2.1. Collective management
- 8.2.2.2. Task allocation
- 8.2.2.3. Ability of swarm to split and merge/ change configuration
- 8.2.2.4. Self healing
- 8.2.2.5. Endurance of swarm
- 8.2.2.6. Operational height of swarm
- 8.2.2.7. Launch and recovery characteristics

8.2.3. UAS Flight Characteristics

- 8.2.3.1. Type of UAS
- 8.2.3.2. Flight stabilization
- 8.2.3.3. Localisation and navigation
- 8.2.3.4. Collision avoidance
- 8.2.3.5. Power and energy management
- 8.2.3.6. Return home (default and in failure)

8.2.4. **Communication**

- 8.2.4.1. LOS/ Any other solution
- 8.2.4.2. Distance projected and achieved
- 8.2.4.3. Downlink and recording
- 8.2.4.4. Back-up data link
- 8.2.4.5. Security features/ encryption

8.2.5. **Maintenance Aspects**

- 8.2.5.1. Concept of maintenance
- 8.2.5.2. Ease of repair/ rectification
- 8.2.5.3. Cost effectiveness

8.2.6. **Miscellaneous**

- 8.2.6.1. Operation by night
- 8.2.6.2. Level of indigenisation for sub-systems and components
- 8.2.6.3. Manpower required for operation
- 8.2.6.4. Operation in GNSS denied environment

9. **Award Administration Information**

9.1. **Phase-0**

Development Funds : Nil

9.2. **Phase-I**

Development Funds : Nil

9.3. **Phase-II**

Development Funds : Up to Rs. 30 Lakh (Rupees Thirty Lakh Only) per team

(Development fund will be provided to up to ten shortlisted participants/ teams who meet a specific level of the scenario)

benchmark as assessed by Committee of Experts. The amount will be reimbursed based on the invoices submitted by the participants/ teams and scrutinised by concerned Accounts Department of the IAF).

Award : Certificate of participation

9.4. Phase-III

Development Funds : Up to Rs. 60 Lakh (Rupees Sixty Lakh Only) per team

(Progressive development fund will be provided to up to five finalists who meet a specific level of the scenario benchmark as assessed by Committee of Experts. The amount will be reimbursed based on the invoices submitted by the participants/ teams and scrutinised by concerned Accounts Department of the IAF).

9.5. Phase-IV

Development Funds : Up to Rs. 1.4 Crore (Rupees One Crore and Forty Lakh) per team

(Development funds will be provided to up to five finalists who meet a specific level of the scenario benchmark as assessed by Committee of Experts. The amount will be reimbursed based on the invoices submitted by the participants/ teams and scrutinised by concerned Accounts Department of the IAF).

Award : Rs. 30 Lakh (Rupees Thirty Lakh Only) to the winner.

Rs. 20 Lakh (Rupees Twenty Lakh Only) to the second position.

Rs. 10 Lakh (Rupees Ten Lakh Only) to the third position.

Certificate of participation to all finalists.

10. Terms and Conditions for Participation

10.1. Only Indian citizens/ entities/ educational institutions/ registered organizations/ companies are allowed to participate.

10.2. All teams need to register by filling up the registration form given on the website with requisite documents. Incomplete forms will not be accepted and would entail the team not being shortlisted for participation.

10.3. A write up with approximately 500 words elaborating the proposed solution along with supporting documentation i.e. any links to video/ presentation is a mandatory requirement for the selection process.

10.4. All participants shall abide by the timelines as given in the website. Non-adherence would entail disqualification of the team.

10.5. The teams would be solely responsible for the safety and security of the equipment.

10.6. The expenditure on travel including transportation of equipment, boarding and lodging shall be borne by the individual participants/ teams. IAF is not liable to pay for any breakages/ damages/ compensation for injury (to self or other personnel) during the complete course of the competition.

10.7. IAF reserves the right to select all, some, one or none of the proposals received in response to this solicitation for negotiation and to make awards without discussion with proposer.

10.8. Participants will be liable for any damage/ collateral damage/ injuries that would occur to any third party/ team members/ participants/ public at demonstration area during the competition while flying their respective UAS.

10.9. The participants shall not publish any research paper without prior permission and vetting from the Air HQ, resulting from the proposed effort as the same is likely to disclose the performance characteristics of military systems or manufacturing technologies that are unique, critical to defence and have bearing on security of the state.

10.10. The decision of Committee of Experts nominated by IAF for the evaluation during the competition will be final and binding on all participants. It will not be subjected to any dispute.

10.11. All suits or legal proceedings of any kind against the decision shall be instituted in the appropriate court(s) in Delhi or New Delhi notwithstanding the location of the competition.

10.12. No suits or legal proceedings of any kind shall be instituted against the decision unless a notice in writing has been delivered to IAF stipulating the nature of claims, cause of action, relief sought, name, registration number and address of the person and a period of two months has expired thereafter.

10.13. The participants shall strictly adhere to all security norms/ restrictions as instructed by IAF from time to time.

10.14. IAF reserves the right to accept proposals in their entirety or to select only portions of proposals for award and subsequent co-production with IAF.

10.15. IAF reserves the right to request further and/ or additional, documentation at any stage of the competition.

10.16. IAF reserves the right to move proposals for award consideration, should the team(s) fail to reach agreement on award terms, conditions within a reasonable time, or the team(s) fail to provide requested additional information in the timelines specified by IAF.

10.17. For the publicity/ advertising of the given competition, the IAF might use the provided documentation. The applicants can opt out of the above but shall need to state the same at the time of providing the information to IAF.

10.18. IAF reserves the right to cancel or reschedule the event without any notification. Under such circumstances, the prospective participants are not entitled for any compensation or relief for the loss of any kind they may suffer.

10.19. IAF holds the right to make any changes and amendments to these terms and conditions at any time without giving any prior knowledge whatsoever.

10.20. Details of resources employed, bills/ invoices of the equipment, sub-systems, components and proprietary algorithms are to be provided by the participants to IAF for the purpose of calculating the indigenous content and the financial outlay.

11. **Competition Contacts**

Address:

Project Officer, Mehar Baba Competition-3
Directorate of Operations (Remote)
Room No. 549, Air Headquarters (Vayu Bhawan)
Motilal Nehru Marg, New Delhi – 110 106

Contact Details:

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011 – 23063674

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Webpage: https://indianairforce.nic.in/mehar_baba/

Point of Contact: Group Captain Vijay Vinchurkar

APPENDIX

(Refers to Para 3.3 of Vision Document for MBC-3)

BROAD PARAMETERS: MBC-3

1. **Competition.** At the end of the competition, the participants are expected to showcase a collaborative solution with swarm of Unmanned Aerial Systems (UAS) working as an airborne radar capable of detecting, tracking and reporting aerial targets in a contested environment and provide their accurate location at a centralised monitoring station.
2. **Broad Parameters for Competition.** The broad parameters including technical specifications, operational requirements and desirable features for the competition are listed below:-
 - 2.1. The solution should comprise a swarm of minimum five UAS which are runway independent (preferably VTOL) or capable of operating from short paved/ unpaved surfaces.
 - 2.2. The swarm should be able to autonomously manoeuvre and also be controlled from the ground.
 - 2.3. Radar detection range of swarm: 2 to 5 km.
 - 2.4. Detect, track and report targets with a Radar Cross Section (RCS) of 0.3 m^2 at 2 km or better.
 - 2.5. Detect, track and report targets having velocity 10 to 40 m/s (36 to 144 kmph).
 - 2.6. Radar Field of View for the swarm should be 90 deg (minimum) at a given time with capability to provide radar coverage all around (360 deg) on demand.
 - 2.7. Revisit time: Below 10 sec.
 - 2.8. Endurance of swarm: 30 minutes (minimum).
 - 2.9. Operational time of radar detection of swarm: 15 minutes (minimum).
 - 2.10. Operational height of swarm: Minimum 500 m (above ground level) and 2 km (AMSL).
 - 2.11. Detect and track multiple targets (five targets or more) in the airspace defined by:-
 - Range: 2 to 5 km
 - Azimuth: +/- 45 degree
 - Within a height band of +/- 500 m with respect to average height of the swarm
 - 2.12. Range resolution: 120 m; Resolution in azimuth and elevation should be such that it is able to provide good multi-target resolvability.
 - 2.13. Combine inputs from all UAS and demonstrate a consolidated near real-time Air Situation Picture (ASP) on a single screen with recording facility.

- 2.14. Graceful degradation in case one or more UAS are not available for operation.
- 2.15. Self healing capability of the swarm
- 2.16. Preferably, employ encrypted data link (desirable characteristic).
- 2.17. Operations by day and night, and light wind conditions (up to 10 knots).
- 2.18. Capable of returning home automatically in case of unserviceability, loss of data link, no report scenario or any other eventuality.
- 2.19. Different modes of operation including detection and tracking by the swarm from one position, and detection while the swarm is moving.
- 2.20. Ability of the swarm to split and merge on demand (may be in a different configuration).
- 2.21. Ability of the UAS and sensor both to operate in GNSS denied environment (desirable characteristic).
- 2.22. The design should afford accessibility to facilitate quick replacement of faulty assemblies/ sub-assemblies/ modules in-situ in field conditions by user.
- 2.23. Ability to operate with minimum manpower.
- 2.24. Ability to operate efficiently in Indian atmospheric conditions.
- 2.25. Level of indigenisation for sub-systems and components should be minimum 50 % considering mission-criticality, safety-criticality and security-criticality. Usage of equipment, components, sub-systems banned by Govt. of India is strictly prohibited.